



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

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Practical No.: 6

Aim –

Use string functions and regex for email validation, data extraction, and text analysis.

Software/Tools Required –

VSCode, Chrome, HTML5, CSS3, Javascript (ES6)

Experiment Program Code –

index.html

```
<!DOCTYPE html>
<html>
  <head>
    <title>
      String Methods and Regular Expressions
    </title>

    <style>
      body {
        font-family: Arial, sans-serif;
        margin: 40px;
      }

      h1 {
        text-align: center;
      }

      .box {
        border: 1px solid #ccc;
        padding: 20px;
        margin-top: 20px;
        border-radius: 8px;
      }

      input, textarea, button {
        padding: 10px;
        margin: 5px 0;
        width: 90%;
      }

      button {
        width: 150px;
        cursor: pointer;
      }
    </style>
  </head>
  <body>
    <h1>String Methods and Regular Expressions</h1>
  </body>
</html>
```

```

    #output {
      margin-top: 20px;
      padding: 15px;
      background-color: #f5f5f5;
    }
  </style>
</head>

<body>
  <h1>String Methods & Regular Expressions</h1>
  <div class="box">
    <label><b>Enter a paragraph:</b></label> <br>

    <textarea id="paragraph" rows="5" >Javascript is a powerful programming language. It is widely used for web
development.
    </textarea>
    <br>

    <label><b>Enter Email:</b></label> <br>
    <input type="text" id="email" placeholder="Enter your email"> <br>

    <button onclick="processString()">Process</button>
    <div id="output"></div>
  </div>

  <script>
    function processString() {
      let paragraph = document.getElementById("paragraph").value;
      let email = document.getElementById("email").value;

      let words = paragraph.split(/\s+/);

      let vowels = paragraph.match(/[aeiou]/gi);
      let vowelCount = vowels ? vowels.length : 0;

      let replacedParagraph = paragraph.replace(
        /JavaScript/gi,
        "JavaScript Programming"
      );

      let searchWord = "powerful";
      let position = paragraph.indexOf(searchWord);

      let emailRegex =
        /^([a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,})$/;

      let emailResult;

      if(emailRegex.test(email)) {
        emailResult = "Valid Email Address";
      } else {
        emailResult = "Invalid Email Address";
      }

      let emailText =
        "For queries contact student@example.com or admin@college.edu";

      let extractedEmails = emailText.match(
        /[a-zA-Z0-9._%+-]+@[a-zA-z0-9.-]+\.[a-zA-Z]{2,}/g
      );

      let reeversedParagraph = paragraph.split("").reverse().join("");

      document.getElementById("output").innerHTML = `
      <h3>1. Original Paragraph</h3>
      <p>${paragraph}</p>

```

```

<h3>2. split() - Words</h3>
<p>${words.join(", ")} </p>

<h3>3. match() - Vowels</h3>
<p>${vowels ? vowels.join(", ") : "No vowels found"}</p>

<h3>4. Vowel Count</h3>
<p>Total number of vowels: <b>${vowelCount}</b></p>

<h3>5. replace() - Replaced Text</h3>
<p>${replacedParagraph}</p>

<h3>6. indexOf() - Search Word</h3>
<p>Position of "<b>${searchWord}</b>";
<b>${position}</b>
</p>

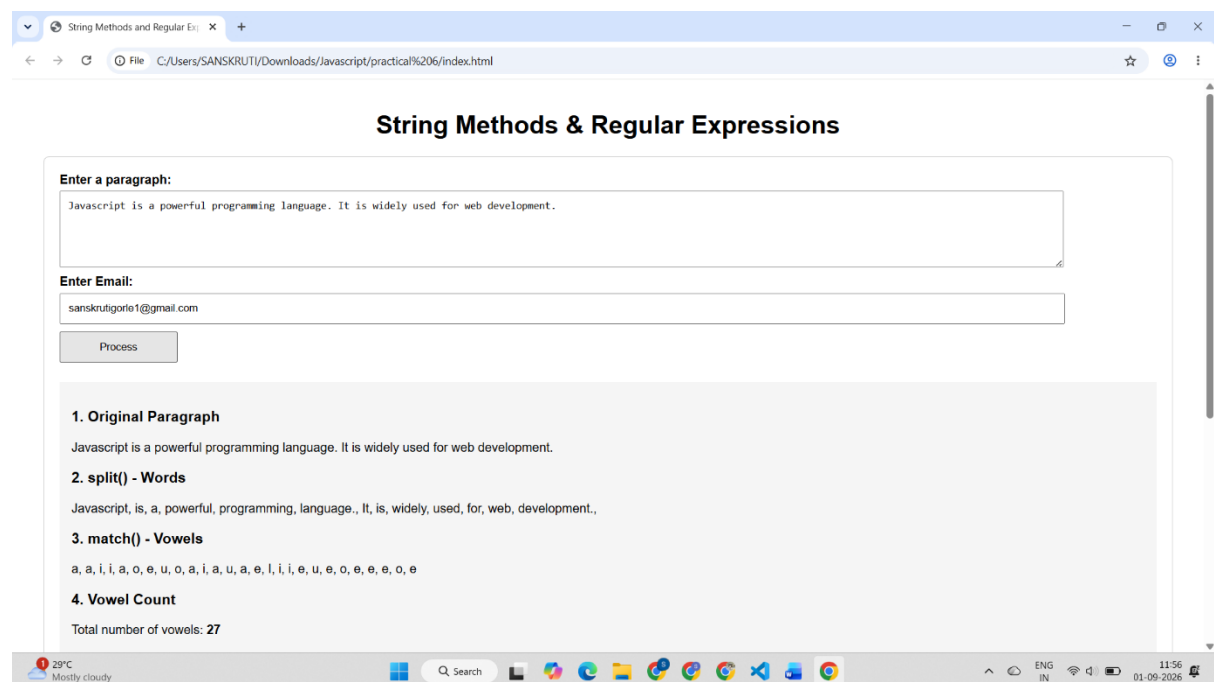
<h3>7. Email Validation Using Regex</h3>
<p>
    Email: <b>${email}</b><br>
    Result: <b>${emailResult}</b>
</p>

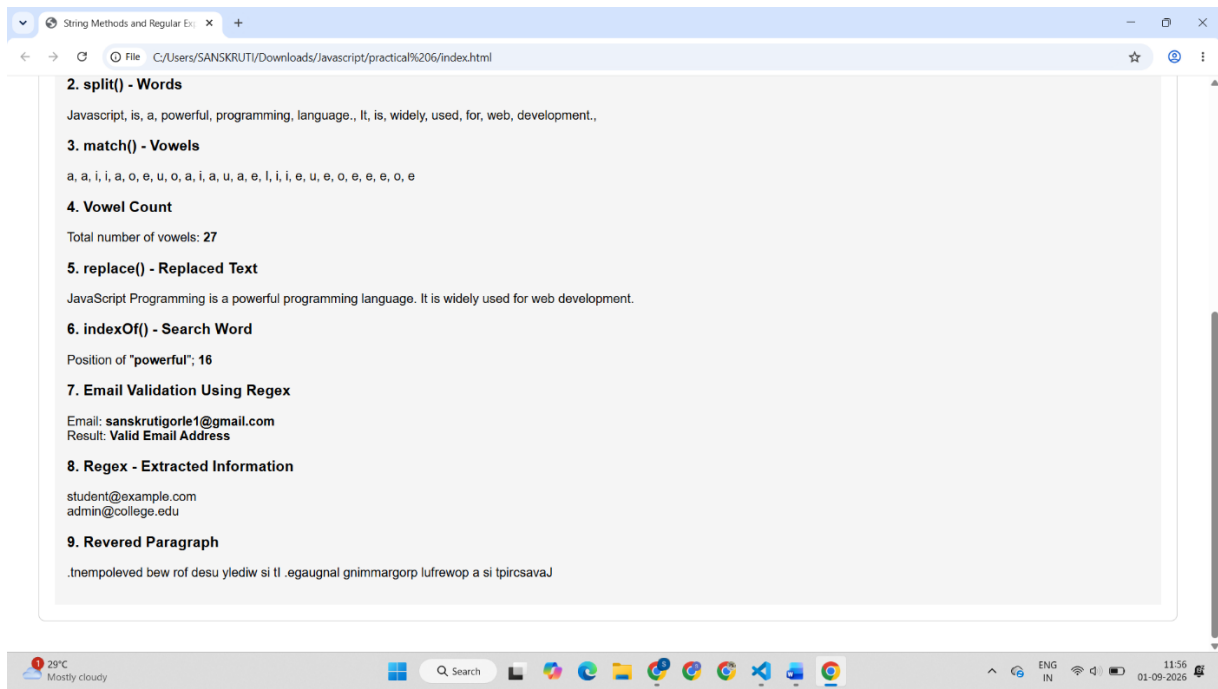
<h3>8. Regex - Extracted Information</h3>
<p>
    ${extractedEmails ? extractedEmails.join("<br>") : "No email address found"}
</p>

<h3>9. Revered Paragraph</h3>
<p>${reeversedParagraph}</p>
    `;
    }
</script>
</body>
</html>

```

Output –





Case Study Title –

To reverse a string and count the number of vowels in a given string or paragraph.

Case Study Program Code –

index.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>StringLab | JavaScript Practical</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <header class="hero">

    <nav>
      <div class="brand">
        <div class="brand-icon">JS</div>
        <span>StringLab</span>
      </div>

      <div class="nav-links">
        <a href="#input">Playground</a>
        <a href="#case-studies">Case Studies</a>
      </div>
    </nav>
  </header>
</body>
</html>
```


</div>

</div>

</section>

<section class="dark-section" id="case-studies">

<div class="section dark-inner">

<div class="section-title dark-title">

<div class="number">

01

</div>

<div>

CASE STUDIES

<h2>

String Operations

</h2>

<p>

Reverse a string and count the number
of vowels in a paragraph.

</p>

</div>

</div>

<div class="analysis-grid">

<div class="analysis-card">

<div class="analysis-icon">

↔

</div>

CASE STUDY 01

<h3>

Reverse a String

</h3>

<p>

Reverse every character of the
entered string using JavaScript.

</p>

<div class="analysis-output">

REVERSED STRING

<div id="reverseOutput">

—

</div>

```
</div>

</div>

<div class="analysis-card">

  <div class="analysis-icon">
    Aa
  </div>

  <span class="analysis-label">
    CASE STUDY 02
  </span>

  <h3>
    Count Vowels
  </h3>

  <p>
    Count the number of vowels present
    in the entered paragraph.
  </p>

  <div class="vowel-stats">

    <div class="stat">

      <span>
        VOWELS
      </span>

      <strong id="vowelOutput">
        0
      </strong>

    </div>

    <div class="stat">

      <span>
        CHARACTERS
      </span>

      <strong id="totalOutput">
        0
      </strong>

    </div>

  </div>

</div>

</div>

<div class="summary">

  <div>
    <span>
      OPERATIONS COMPLETED
    </span>

    <strong id="operationCount">
      0 / 2
    </strong>
  </div>

</div>
```

```

        <div class="summary-line"></div>

        <p>
            Both case studies use the same input string.
        </p>

    </div>

</div>

</section>

</main>

<footer>

    <div class="footer-inner">

        <div class="footer-brand">

            <div class="brand-icon">
                JS
            </div>

            <div>
                <strong>
                    StringLab
                </strong>

                <p>
                    JavaScript Practical
                </p>
            </div>

        </div>

        <span>
            HTML • CSS • JavaScript
        </span>

    </div>

</footer>

<script src="script.js"></script>

</body>

</html>

```

style.css

```

@import
url('https://fonts.googleapis.com/css2?family=DM+Sans:wght@400;500;600;700&family=Space+Grotesk:wght@500;600;700&display=swap');

* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

```



```
html {
  scroll-behavior: smooth;
}

body {
  font-family: "DM Sans", sans-serif;
  background: #f7f7fa;
  color: #17181d;
  line-height: 1.6;
}

.hero {
  min-height: 600px;
  background:
    radial-gradient(
      circle at 80% 20%,
      rgba(105, 87, 232, 0.28),
      transparent 30%
    ),
    radial-gradient(
      circle at 10% 90%,
      rgba(156, 143, 255, 0.12),
      transparent 30%
    ),
    #101116;
  color: white;
  overflow: hidden;
  position: relative;
}

.hero::before {
  content: "";
  position: absolute;
  width: 520px;
  height: 520px;
  border: 1px solid rgba(255,255,255,0.06);
  border-radius: 50%;
  right: -220px;
  top: -180px;
}

.hero::after {
  content: "";
  position: absolute;
  width: 330px;
  height: 330px;
  border: 1px solid rgba(255,255,255,0.05);
  border-radius: 50%;
  left: -180px;
  bottom: -170px;
}

.nav {
  max-width: 1180px;
  margin: auto;
  padding: 28px 30px;
  display: flex;
  justify-content: space-between;
  align-items: center;
  position: relative;
  z-index: 2;
}

.brand {
  display: flex;
  align-items: center;
  gap: 12px;
```

```
font-family: "Space Grotesk";
font-weight: 600;
font-size: 18px;
}

.brand-icon {
width: 40px;
height: 40px;
display: grid;
place-items: center;
background: #6957e8;
border-radius: 11px;
font-size: 12px;
font-weight: 700;
}

.nav-links {
display: flex;
gap: 30px;
}

.nav-links a {
text-decoration: none;
color: #999ba5;
font-size: 13px;
transition: 0.2s;
}

.nav-links a:hover {
color: white;
}

.hero-content {
max-width: 900px;
margin: 110px auto;
padding: 0 30px;
position: relative;
z-index: 2;
}

.badge {
display: inline-block;
padding: 7px 13px;
border: 1px solid rgba(255,255,255,0.15);
border-radius: 30px;
color: #bec5ff;
background: rgba(255,255,255,0.035);
font-size: 11px;
font-weight: 600;
letter-spacing: 1.5px;
text-transform: uppercase;
margin-bottom: 22px;
}

.hero h1 {
font-family: "Space Grotesk";
font-size: clamp(48px, 7vw, 80px);
line-height: 1;
letter-spacing: -3px;
}

.hero h1 span {
color: #9c8fff;
}

.hero p {
max-width: 600px;
```

```
    color: #9b9da6;
    margin: 25px 0 32px;
    font-size: 16px;
}

.hero-button {
    display: inline-block;
    padding: 13px 21px;
    background: #6957e8;
    color: white;
    text-decoration: none;
    border-radius: 10px;
    font-size: 13px;
    font-weight: 600;
    transition: 0.2s;
}

.hero-button:hover {
    background: #5745d3;
    transform: translateY(-2px);
}

.section {
    max-width: 1180px;
    margin: auto;
    padding: 90px 30px;
}

.section-label {
    color: #6957e8;
    font-size: 10px;
    font-weight: 700;
    letter-spacing: 2px;
}

.input-header {
    margin-top: 12px;
    display: flex;
    justify-content: space-between;
    align-items: end;
    gap: 20px;
}

.input-header h2 {
    font-family: "Space Grotesk";
    font-size: 38px;
    letter-spacing: -1px;
}

.input-header p {
    color: #737680;
    font-size: 14px;
    margin-top: 5px;
    max-width: 620px;
}

.input-badge {
    padding: 7px 12px;
    background: #ecea9f;
    color: #6957e8;
    border-radius: 7px;
    font-size: 10px;
    font-weight: 700;
    letter-spacing: 1px;
}

.input-card {
```

```
background: white;
border: 1px solid #e6e6eb;
border-radius: 20px;
padding: 25px;
margin-top: 35px;
box-shadow: 0 20px 60px rgba(25,25,50,0.07);
}

.input-card textarea {
width: 100%;
height: 145px;
resize: vertical;
border: none;
outline: none;
background: #fafafd;
border-radius: 13px;
padding: 18px;
font-family: inherit;
font-size: 15px;
color: #17181d;
}

.input-card textarea:focus {
box-shadow: inset 0 0 0 1px #6957e8;
}

.input-card textarea::placeholder {
color: #a0a1aa;
}

.input-footer {
display: flex;
justify-content: space-between;
align-items: center;
margin-top: 18px;
}

.input-footer span {
color: #737680;
font-size: 12px;
}

button {
border: none;
cursor: pointer;
color: white;
background: #6957e8;
padding: 13px 21px;
border-radius: 10px;
font-family: inherit;
font-weight: 600;
font-size: 13px;
transition: 0.2s;
}

button:hover {
background: #5745d3;
transform: translateY(-1px);
}

.dark-section {
background: #101116;
color: white;
}

.dark-inner {
padding-top: 90px;
}
```

```
padding-bottom: 90px;
}

.section-title {
  display: grid;
  grid-template-columns: 65px 1fr;
  gap: 20px;
  margin-bottom: 42px;
}

.number {
  color: #6957e8;
  font-family: "Space Grotesk";
  font-size: 14px;
  font-weight: 700;
  padding-top: 7px;
}

.eyebrow {
  color: #6957e8;
  font-size: 10px;
  font-weight: 700;
  letter-spacing: 2px;
}

.section-title h2 {
  font-family: "Space Grotesk";
  font-size: 42px;
  letter-spacing: -1.5px;
  margin: 5px 0;
}

.section-title p {
  color: #737680;
  font-size: 14px;
  max-width: 650px;
}

.dark-title h2 {
  color: white;
}

.dark-title p {
  color: #858792;
}

.analysis-grid {
  display: grid;
  grid-template-columns: 1fr 1fr;
  gap: 18px;
}

.analysis-card {
  background: #191a21;
  border: 1px solid #292a32;
  border-radius: 20px;
  padding: 30px;
}

.analysis-icon {
  width: 45px;
  height: 45px;
  display: grid;
  place-items: center;
  background: rgba(105,87,232,0.15);
  color: #a398ff;
  border-radius: 12px;
```

```
font-weight: 700;
margin-bottom: 22px;
}

.analysis-label {
  color: #8176e9;
  font-size: 9px;
  letter-spacing: 1.5px;
  font-weight: 700;
}

.analysis-card h3 {
  font-family: "Space Grotesk";
  font-size: 25px;
  margin: 6px 0;
}

.analysis-card > p {
  color: #81838d;
  font-size: 13px;
  max-width: 420px;
}

.analysis-output {
  margin-top: 25px;
  background: #111218;
  border-radius: 11px;
  padding: 15px;
}

.analysis-output > span {
  display: block;
  color: #686a75;
  font-size: 9px;
  letter-spacing: 1.5px;
  font-weight: 700;
  margin-bottom: 8px;
}

#reverseOutput {
  color: #c2bcff;
  font-size: 13px;
  word-break: break-word;
  min-height: 22px;
}

.vowel-stats {
  display: grid;
  grid-template-columns: 1fr 1fr;
  gap: 10px;
  margin-top: 25px;
}

.stat {
  background: #111218;
  padding: 17px;
  border-radius: 11px;
}

.stat span {
  display: block;
  color: #686a75;
  font-size: 9px;
  letter-spacing: 1.5px;
  font-weight: 700;
}
```

```
.stat strong {
  display: block;
  font-family: "Space Grotesk";
  font-size: 28px;
  margin-top: 3px;
  color: #c2bcff;
}
```

```
.summary {
  margin-top: 25px;
  padding: 20px 0;
  display: flex;
  align-items: center;
  gap: 25px;
  border-top: 1px solid #25262d;
}
```

```
.summary span {
  color: #666873;
  font-size: 9px;
  letter-spacing: 1.5px;
  font-weight: 700;
  display: block;
}
```

```
.summary strong {
  font-family: "Space Grotesk";
  font-size: 18px;
  color: #c3bdff;
}
```

```
.summary-line {
  width: 1px;
  height: 30px;
  background: #292a31;
}
```

```
.summary p {
  color: #666873;
  font-size: 11px;
}
```

```
footer {
  background: #0b0c10;
  color: white;
}
```

```
.footer-inner {
  max-width: 1180px;
  margin: auto;
  padding: 30px;
  display: flex;
  justify-content: space-between;
  align-items: center;
}
```

```
.footer-brand {
  display: flex;
  align-items: center;
  gap: 12px;
}
```

```
.footer-brand strong {
  font-family: "Space Grotesk";
}
```

```
.footer-brand p {
```

```
color: #666873;
font-size: 10px;
}

.footer-inner > span {
color: #555761;
font-size: 10px;
}

@media (max-width: 800px) {

.nav-links {
display: none;
}

.hero-content {
margin-top: 80px;
}

.section {
padding: 70px 20px;
}

.input-header {
align-items: flex-start;
flex-direction: column;
}

.section-title {
grid-template-columns: 1fr;
gap: 5px;
}

.section-title h2 {
font-size: 34px;
}

.analysis-grid {
grid-template-columns: 1fr;
}
}

@media (max-width: 520px) {

.hero h1 {
font-size: 48px;
letter-spacing: -2px;
}

.input-footer {
align-items: stretch;
flex-direction: column;
gap: 15px;
}

.input-footer button {
width: 100%;
}

.vowel-stats {
grid-template-columns: 1fr;
}

.summary {
align-items: flex-start;
flex-direction: column;
gap: 10px;
}
```



```

}

.summary-line {
  width: 100%;
  height: 1px;
}

.footer-inner {
  align-items: flex-start;
  flex-direction: column;
  gap: 15px;
}
}

```

script.js

```

function processString() {

  let paragraph = document.getElementById("mainInput").value;

  if (paragraph.trim() === "") {

    alert("Please enter a string or paragraph.");

    return;
  }

  let reversedParagraph =
    paragraph.split("").reverse().join("");

  let vowels =
    paragraph.match(/[aeiou]/gi);

  let vowelCount =
    vowels ? vowels.length : 0;

  document.getElementById("reverseOutput").innerHTML =
    reversedParagraph;

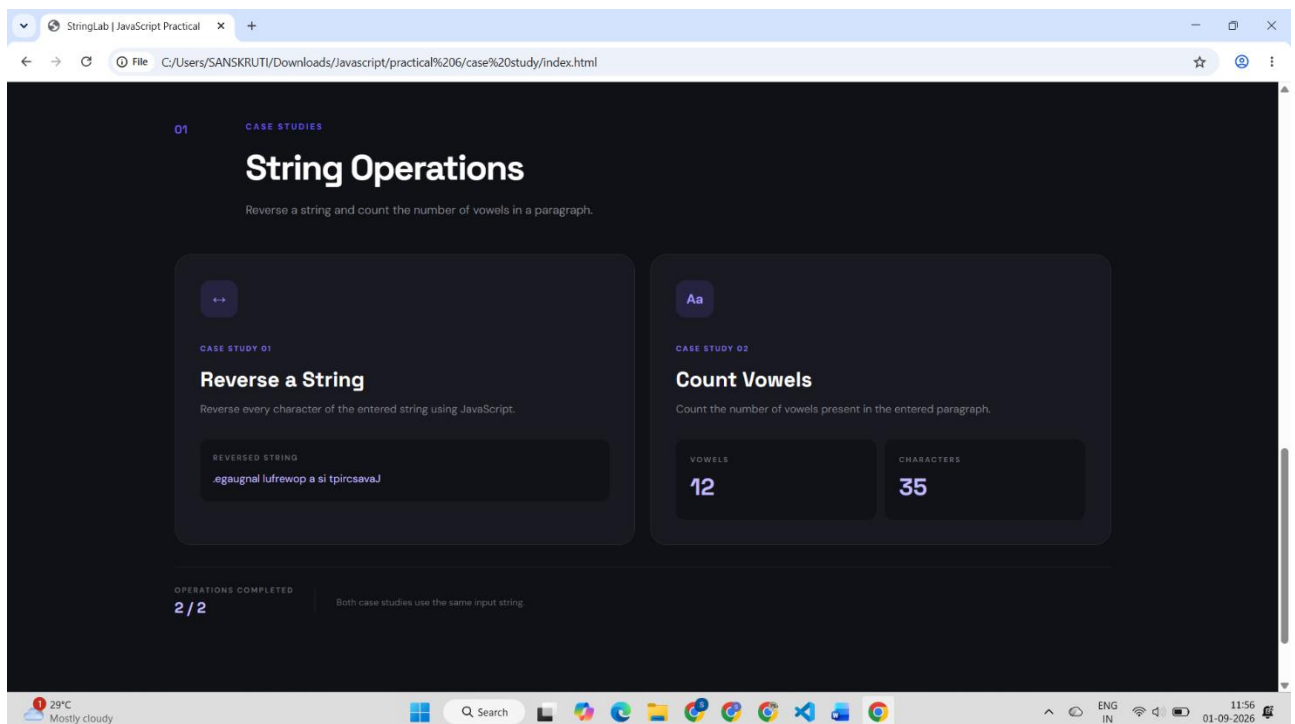
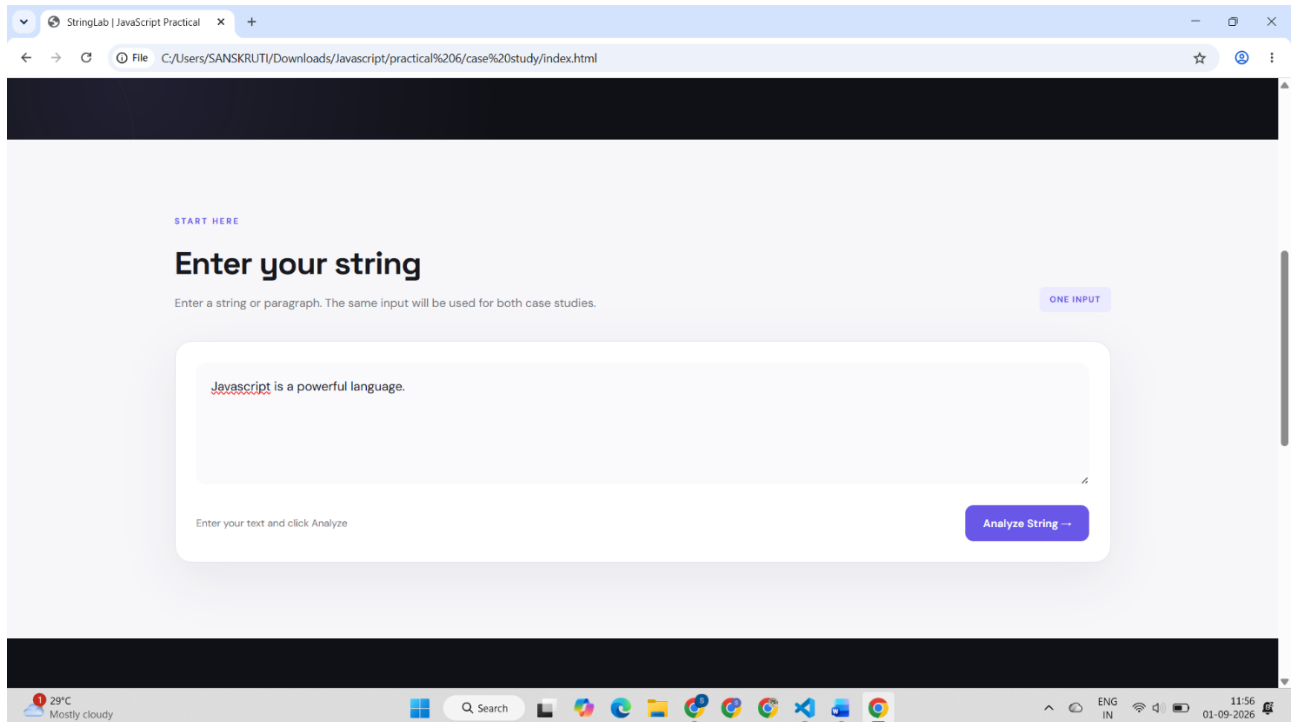
  document.getElementById("vowelOutput").innerHTML =
    vowelCount;

  document.getElementById("totalOutput").innerHTML =
    paragraph.length;

  document.getElementById("operationCount").innerHTML =
    "2 / 2";
}

```

Output –



Conclusion –

The experiment was successfully performed to study various JavaScript string functions and regular expressions. Different string operations such as `split()`, `match()`, `replace()`, and `indexOf()` were used for text analysis. Regular expressions were also implemented for email validation and extraction of email addresses.

In the case study, a string was successfully reversed and the number of vowels and total characters were calculated. Thus, the practical helped in understanding string manipulation and the application of regular expressions in JavaScript.